Neural Differentiation of Primary and Secondary Reward Processing in Addiction: A Systematic Review

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Introduction

Addiction: Value-based Decision Making

A chronic brain disease involving disruptions in reward processing and motivational systems (Volkow et al.,2017)

Reward: "liking" & "wanting" (Sescousse et al., 2013)

- Primary Reward (e.g., food, sex, drugs)
- Secondary Reward (e.g., money, social status)

Gap: The need for an integrated perspective

Although these reward types rely on overlapping neural circuits, they have typically been studied separately.

Research Question

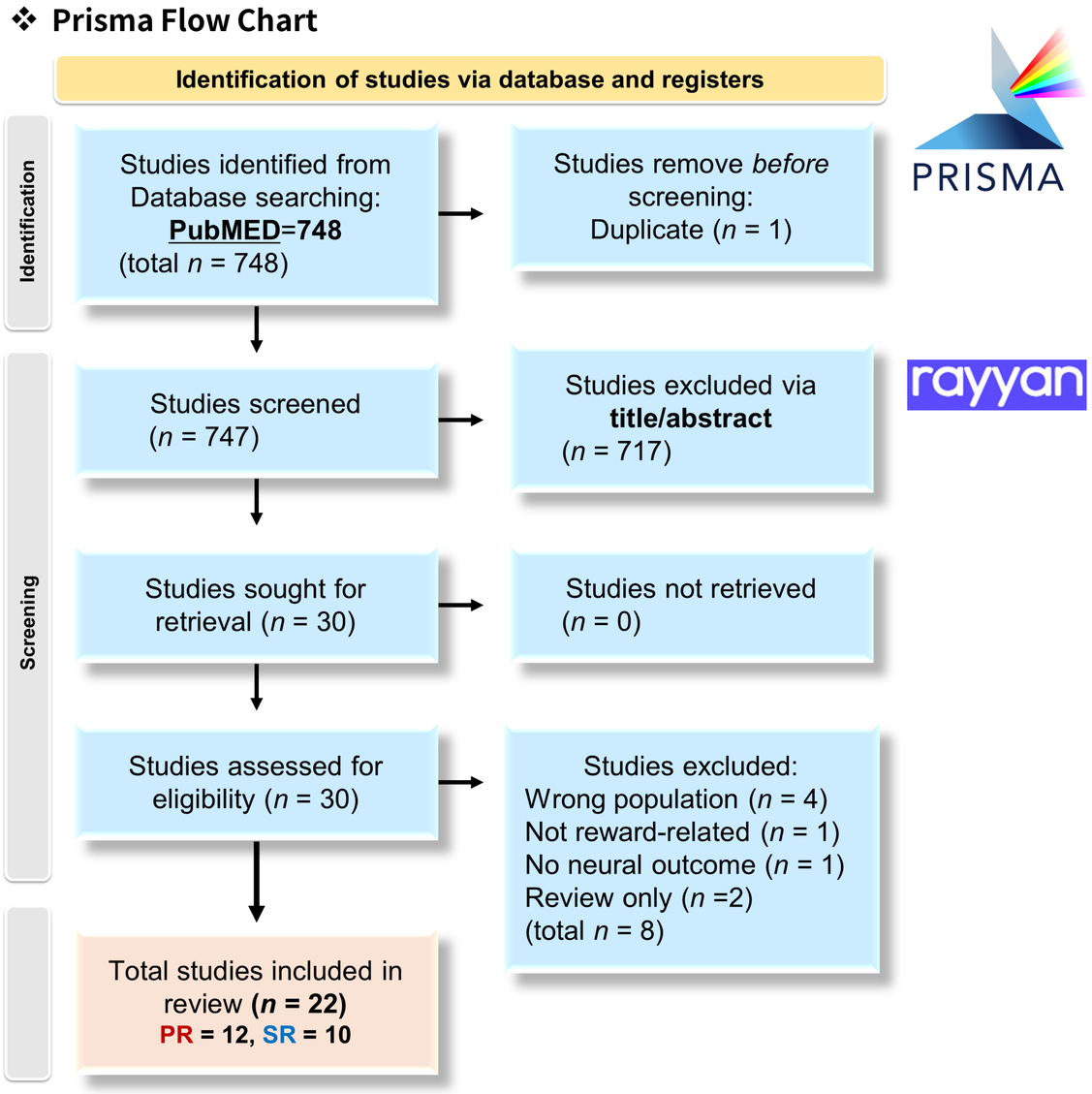
What are the common and differential neural responses to primary(food, sex, drugs) and secondary(money) reward stimuli within the reward circuit among individuals with substance or behavioral addictions?

Objectives

- Conduct a systematic review following PRISMA 2020 (Page et al., 2021)
- Compare neural responses to primary and secondary rewards by integrating findings across different types of addictions
- Identify gaps and suggest future research directions

Methods

❖ **Prisma Flow Chart**



✓ **Inclusion Criteria**

- Participants: individuals with substance use disorder, behavioral addiction, or healthy controls
- Neural outcomes reported: fMRI, PET
- Experimental task design involving reward stimuli

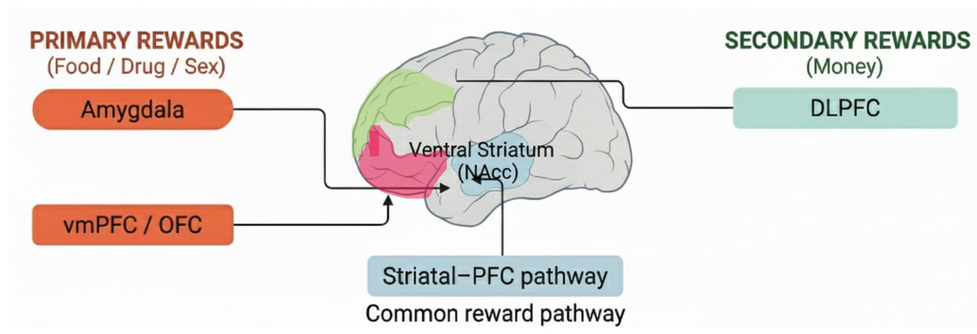
X **Exclusion Criteria**

- Animal studies only
- No neural outcomes
- Incorrect reward categorization
- Excluded publication types

Results

❖ **Table. Summary of Shared and Distinct Neural Signatures Across Reward Types**

Neural Domain	Primary Rewards (Food, Drug, Sex)	Secondary Rewards (Money)	Interpretation
Striatum (VS/ NAcc)	**Strong cue reactivity**; hyperactivation with drug/ food cues	**Strong activation** during monetary anticipation & outcomes	**Shared core reward mechanism**
Prefrontal Cortex (vmPFC / OFC)	**vmPFC & OFC** Valuation-related activation; linked to craving	OFC & **DLPFC** recruited during monetary decision-making	**Common involvement, Different functions**
Amygdala	**Robust activation** during cue exposure and food/drug anticipation	**Weak or inverse activation**; correlated negatively with severity	**Primary = emotional salience Secondary = evaluative**
Connectivity	Impaired **ACC–striatal & limbic connectivity** (FA)	Disrupted **dorsal–ventral striatal coupling** (GD)	**Different connectivity disruption patterns**



Conclusions

- Primary and Secondary rewards share **core striatal mechanisms** but differ in **amygdala** and **prefrontal responses**.
- Addiction types show distinct patterns of **prefrontal-striatal connectivity disruption**.
- Future studies should adopt naturalistic reward paradigms to improve ecological validity

References

- Volkow, N. D., Wise, R. A., & Baler, R. (2017). The dopamine motive system: Implications for drug and food addiction. *Nature Reviews Neuroscience, 18*(12), 741–752.
- Sescousse, G., Caldú, X., Segura, B., & Dreher, J. C. (2013). Processing of primary and secondary rewards: A quantitative meta-analysis and review of human functional neuroimaging studies. *Neuroscience & Biobehavioral Reviews, 37*(4), 681–696.
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., ... & Moher, D. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ, 372.*
- *Additional references are provided in a separate paper*